

APPENDIX

Problem Decomposition into SP1 and SP2

1) Social cost minimization subproblem

By the arithmetic-geometric mean inequality, the objective in (11) attains maximum when the following condition holds:

$$\begin{aligned} & (f_{\text{DSO},0}^* - f_{\text{DSO}}) \prod_{j \in \Omega_N, h \in \Omega_{\text{DER}_j}} (f_{\text{agent},0,h,j}^* - f_{\text{agent},h,j}) = \\ & \left\{ \frac{1}{1+n_{\text{agent}}} \left[f_{\text{DSO},0}^* - f_{\text{DSO}} + \sum_{j \in \Omega_N} \sum_{h \in \Omega_{\text{DER}_j}} (f_{\text{agent},0,h,j}^* - f_{\text{agent},h,j}) \right] \right\}^{1+n_{\text{agent}}} \end{aligned} \quad (\text{A1})$$

where n_{agent} denotes the amount of all the DER agents in the system.

Since power trading between the DSO and DER agents is reciprocal as in (10), with equal payment prices and opposite trading quantities, it follows that

$$C_{\text{DSO}}^{\text{tra}} + \sum_{j \in \Omega_N} \sum_{h \in \Omega_{\text{DER}_j}} C_{\text{agent},h,j}^{\text{tra}} = 0 \quad (\text{A2})$$

Substituting (A2) into (A1), the objective function in (11) can be equivalently written as

$$\max -C_{\text{DSO}}^{\text{op}} - \sum_{j \in \Omega_N} \sum_{h \in \Omega_{\text{DER}_j}} C_{\text{agent},h,j}^{\text{op}} \quad (\text{A3})$$

Therefore, maximizing the objective in (11) is equivalent to a social cost minimization problem formulated as SP1 in (A4), which is an MILP problem.

$$(\text{SP1}): \min C_{\text{DSO}}^{\text{op}} + \sum_{j \in \Omega_N} \sum_{h \in \Omega_{\text{DER}_j}} C_{\text{agent},h,j}^{\text{op}} \quad (\text{A4})$$

$$\text{s.t. (2),(4)-(8),(10)}$$

2) Payment bargaining subproblem

SP1 is solved to obtain the operating strategies of the DSO and DER agents, as well as the traded power. Substituting the resulting optimal solutions into (10)-(11) leads to the following payment bargaining subproblem.

$$(\text{SP2}): \max_{\{\lambda_{h,j,t}^{\text{ex},\phi}\}} \left\{ \ln \left[f_{\text{DSO},0}^* - C_{\text{DSO}}^{\text{op}*} - C_{\text{DSO}}^{\text{tra}} (\lambda_{h,j,t}^{\text{ex},\phi}) \right] + \sum_{j \in \Omega_N} \sum_{h \in \Omega_{\text{DER}_j}} \ln \left[f_{\text{agent},0,h,j}^* - C_{\text{agent},h,j}^{\text{op}*} - C_{\text{agent},h,j}^{\text{tra}} (\lambda_{h,j,t}^{\text{ex},\phi}) \right] \right\} \quad (\text{A5a})$$

$$f_{\text{DSO},0}^* - f_{\text{DSO}} \geq 0, \quad f_{\text{agent},0,h,j}^* - f_{\text{agent},h,j} \geq 0 \quad (\text{A5b})$$

where SP2 in (A5) is a convex optimization problem, (A5b) ensures that each participant's payoff is no less than in the non-cooperative case.

3) Matrix forms of the two subproblem models

To facilitate the presentation of the models, SP1 and SP2 are reformulated in matrix form as (A6) and (A7), respectively.

$$\min C_{\text{DSO}}^{\text{op}}(\mathbf{x}_{\text{DSO}}) + \sum_{n \in \Omega_{\text{DER}}} C_n^{\text{op}}(\mathbf{x}_n) \quad (\text{A6a})$$

$$\text{s.t. } \mathbf{x}_{\text{DSO}} \in X_{\text{DSO}} \quad (\text{A6b})$$

$$\mathbf{x}_n \in X_n, \forall n \in \Omega_{\text{DER}} \quad (\text{A6c})$$

$$H_{n \rightarrow \text{DSO}} \mathbf{x}_n = -H_{\text{DSO} \rightarrow n} \mathbf{x}_{\text{DSO}}, \forall n \in \Omega_{\text{DER}} \quad (\text{A6d})$$

$$H_{\text{DSO} \rightarrow n} \mathbf{x}_{\text{DSO}} = \mathbf{P}_n^{\text{ex},\Phi}, \forall n \in \Omega_{\text{DER}} \quad (\text{A6e})$$

$$H_{n \rightarrow \text{DSO}} \mathbf{x}_n = \mathbf{P}_n^{\text{ex},\Phi}, \forall n \in \Omega_{\text{DER}} \quad (\text{A6f})$$

where the vector $\mathbf{x}_{\text{DSO}} = \{\mathbf{P}_{\text{sub}}^{\Phi}, \mathbf{Q}_{\text{sub}}^{\Phi}, \mathbf{P}_{ij}^{\Phi}, \mathbf{Q}_{ij}^{\Phi}, \mathbf{P}_{h,j}^{\text{ex},\Phi}, \mathbf{v}_j^{\Phi}\}$ and $\mathbf{x}_n = \{\mathbf{s}_n^{W,\Phi}, \mathbf{v}_n^{W,\Phi}, \mathbf{u}_n^W, \mathbf{P}_{h,j}^{\text{ex},\Phi}, \mathbf{x}_n^W\}$. For simplicity, the time index t is omitted, and $\mathbf{x}_n^W$ denotes the vector of intrinsic variables of all DER units within DER agent n , where each DER agent $n \in \Omega_{\text{DER}}$ corresponds to a unique pair $(j, h) \in$

$(\Omega_N, \Omega_{\text{DER}_j})$. X_{DSO} and X_n denote the feasible sets of $\mathbf{x}_{\text{DSO}}$ and $\mathbf{x}_n$, respectively. Notably, X_n is a non-convex set constructed by the mixed-integer linear constraints. (A6d) denotes the coupling constraints, where $H_{\text{DSO} \rightarrow n}$ and $H_{n \rightarrow \text{DSO}}$ are constant matrices satisfying (A6e)-(A6f).

$$\max U_{\text{DSO}}(\boldsymbol{\lambda}_{\text{DSO}}) + \sum_{n \in \Omega_{\text{DER}}} U_n(\boldsymbol{\lambda}_n) \quad (\text{A7a})$$

$$\text{s.t. } \boldsymbol{\lambda}_{\text{DSO}} \in Y_{\text{DSO}} \quad (\text{A7b})$$

$$\boldsymbol{\lambda}_n \in Y_n, \forall n \in \Omega_{\text{DER}} \quad (\text{A7c})$$

$$\boldsymbol{\lambda}_n = \boldsymbol{\lambda}_{\text{DSO}} = \boldsymbol{\lambda}_n^{\text{ex},\Phi}, \forall n \in \Omega_{\text{DER}} \quad (\text{A7d})$$

where U_{DSO} and U_n represent the objective function components in (A5a) of the DSO and each DER agent, respectively. The vector $\boldsymbol{\lambda}_{\text{DSO}}$ and $\boldsymbol{\lambda}_n$ denote the pricing decision variables of the DSO and DER agents, respectively. Y_{DSO} and Y_n denote the feasible sets of $\boldsymbol{\lambda}_{\text{DSO}}$ and $\boldsymbol{\lambda}_n$, respectively. Each DER agent $n \in \Omega_{\text{DER}}$ corresponds to a unique pair $(j, h) \in (\Omega_N, \Omega_{\text{DER}_j})$. (A7d) denotes the coupling constraints.